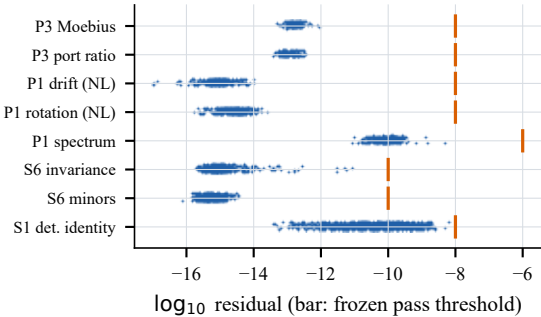
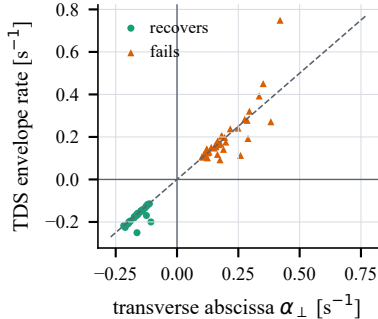


(a) exact identities



(b) nonlinear TDS, 5 MW pulse

(c) crossings closing changes of $\mathcal{H}$ 